6.4 Projective Varieties

6.4.1 Projective Algebraic Sets

Proposition 6.4.1 (Projective Space is Functorial)
Let $\phi : K \to L$ be an injective ring homomorphism Then there is a well-defined injective map

$$\begin{aligned} \mathbb{P}^n(K) &\to \mathbb{P}^n(L) \\ (x_0 : ... : x_n) &\to (\phi(x_0) : ... : \phi(x_n)) \end{aligned}$$

Proof. Evidently the map is well-defined. Suppose that

$$(\phi(x_0) : ... : \phi(x_n)) \sim (\phi(x_0') : ... : \phi(x_n'))$$

Then for some i we have $x_i \neq 0$ and $\phi(x_i) = \lambda\phi(x_i')$ and $\lambda \in L^\star$. Therefore $\phi(x_i') \neq 0 \implies x_i' \neq 0$ and $\lambda = \phi(x_i/x') =: \phi(\mu)$. Similarly $\phi\big(x_j\big) = \phi(\mu)\phi\big(x_j'\big) \implies \phi\big(x_j - \mu x_j'\big) = 0 \implies x_j = \mu x_j'$. ∎

Proposition 6.4.2 (Zero Loci in Projective Space)
*Let $A = k[X_0, ..., X_n]$ be the graded polynomial ring in $(n+1)$-variables over a field k. For a homogenous ideal $\mathfrak{a} \subset k[X_0, ..., X_n]$ and field extension K/k define the **zero-locus***

$$V_+(\mathfrak{a})(K) \coloneqq \{(x_0 : ... : x_n) \in \mathbb{P}^n(K) \mid F(x_0, ..., x_n) = 0 \quad \forall F \in \mathfrak{a}\}$$

Similarly for a subset $Y \subset \mathbb{P}^n(K)$ define

$$I_+(Y) \coloneqq \{F \in k[X_0, ..., X_n] \mid F_d(x_0, ..., x_n) = 0 \quad \forall d \geq 0, (x_0 : ... : x_n) \in Y\}$$

Then

a) $I_+(Y)$ is a homogenous radical ideal and $\sqrt{\langle S \rangle} \subseteq I_+(V_+(S)(K))$
b) I_+ and V_+ are order-reversing
c) $S \subseteq I_+(V_+(S)(K))$ and $Y \subseteq V_+(I_+(Y))(K)$
d) $V_+ \circ I_+ \circ V_+ = V_+$ and $I_+ \circ V_+ \circ I_+ = I_+$
e) $V_+(S) = V_+\Big(\sqrt{\langle S \rangle}\Big)$ and $V_+(\mathfrak{a}) = V_+(\sqrt{\mathfrak{a}})$
f) $\bigcap_i V_+(S_i) = V_+\Big(\bigcup_i S_i\Big)$ and $\bigcap_i V_+(\mathfrak{a}_i) = V_+\Big(\sum_i \mathfrak{a}_i\Big)$
g) $\bigcap_i I_+(W_i) = I_+(\bigcup W_i)$
h) $V_+(\mathfrak{a}) \cup V_+(\mathfrak{b}) = V_+(\mathfrak{a} \cap \mathfrak{b})$
i) $V_+((0)) = \mathbb{P}^n(K)$

*The sets of the form $V_+(\mathfrak{a})(K)$ determine the **Zariski Topology** on $\mathbb{P}^n(K)$. We denote the topological space by $\mathbb{P}^n_k(K)$. The topological closure satisfies the following identities*

$$V_+(I_+(X)) = \overline{X}$$

and

$$I_+(X) = I_+\Big(\overline{X}\Big)$$

There is a form of the Weak Nullstellensatz

$$V_+(\mathfrak{a})\Big(\overline{k}\Big) = \emptyset \iff A_+ \not\subset \sqrt{\mathfrak{a}} \iff \mathfrak{a} \text{ irrelevant}$$

and a form of the Strong Nullstellensatz namely

$$\mathfrak{a} \text{ essential} \implies I_+\Big(V_+(\mathfrak{a})\Big(\overline{k}\Big)\Big) = \sqrt{\mathfrak{a}}$$

Further there is a dual isomorphism

$$\big\{\mathfrak{a} \lhd A \;\big|\; \mathfrak{a} \text{ radical homogenous and } A_+ \not\subset \mathfrak{a}\big\} \underset{I_+}{\overset{V_+}{\rightleftarrows}} \Big\{Z \in \mathbb{P}^n_k\Big(\overline{k}\Big) \;\Big|\; Z \neq \emptyset\Big\}$$

Proof. Let $V_K(S) \coloneqq \big\{(x) \in K^{n+1} \;\big|\; f(x) = 0 \quad \forall s \in S\big\}$ denote the affine zero-locus. Then $V_+(S)(K) = \pi(V_K(S) \setminus \{\mathbf{0}\})$, where $\mathbf{0} = (0, ..., 0)$ is the origin in K^{n+1}.

a) Homogeneity follows from definition. The relation is straightforward.
b) Follows by definition
c) The pair I, V form a Galois connection by (2.1.54) and the relations follow formally by (2.1.50).

© David Rufino CC BY-NC-ND 4.0

d) Follows formally by (2.1.52)
e) Follows from (2.1.53) by considering the closure operators $\sqrt{\langle - \rangle}$ and $\langle - \rangle$.
f) The first equality follows from (2.1.55). The second equality follows from (3.3.29).
g) This follows from (2.1.55).
h) Then using the affine case we may deduce

$$V_+(\mathfrak{a} \cap \mathfrak{b}) = \pi(V(\mathfrak{a}) \cup V(\mathfrak{b}) \setminus \mathbf{0}) = \pi(V(\mathfrak{a}) \setminus \mathbf{0}) \cup \pi(V(\mathfrak{b}) \setminus \mathbf{0}) = V_+(\mathfrak{a}) \cup V_+(\mathfrak{b})$$

i) Trivial

Weak Nullstellensatz. Observe that using the affine case $V_{\overline{k}}(A_+) = \{\mathbf{0}\}$ and $I_{\overline{k}}(\mathbf{0}) = A_+$. Therefore

$$V_+(\mathfrak{a})\left(\overline{k}\right) = \emptyset \iff V_{\overline{k}}(\mathfrak{a}) \subset \{\mathbf{0}\} \overset{(3.12.7)}{\iff} A_+ \not\subset \sqrt{\mathfrak{a}}$$

Strong Nullstellensatz. Assume $\mathfrak{a}$ is an essential homogenous ideal then $Y \coloneqq V(\mathfrak{a}) \setminus \{\mathbf{0}\}$ is non-empty and $\overline{Y} = V(\mathfrak{a})$. Because $\mathfrak{a}$ is homogenous then Y is a cone – $x \in Y \implies \lambda x \in Y$ for $\lambda \in \overline{k}^\star$. We claim that $V_+(\mathfrak{a}) = \pi(Y)$ and $I(Y) = I_+(\pi(Y))$. Suppose $F \in I(Y)$ then $F(\lambda x) = 0$ for all $\lambda \in \overline{k}^\star$ and $x \in Y$. As $\overline{k}$ is infinite we deduce that $F_d \in I(Y)$ whence $F \in I_+(\pi(Y))$. The remaining claims are clear. Therefore we may deduce from (6.1.2) and (6.1.13)

$$I_+V_+(\mathfrak{a}) = I_+(\pi(Y)) = I(Y) = I\left(\overline{Y}\right) = I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$$

Topological Closure. By (2.1.52) $V_+ \circ I_+$ is a closure operator with image precisely the closed sets. Therefore by (2.1.41) $V_+ \circ I_+$ is the topological closure (4.1.19). Finally $I_+\left(\overline{X}\right) = I_+(V_+(I_+(X))) = I_+(X)$ using d).

The dual isomorphism follows from the Strong Nullstellensatz. ∎

Definition 6.4.3 (Projective Variety)
For an essential homogenous radical ideal $\mathfrak{a} \lhd k[X_0, ..., X_n]$ we say the family

$$X(K) \coloneqq V_+(\mathfrak{a})(K) \subset \mathbb{P}^n_k(K)$$

*is a **projective variety** and write it as $X(K) \coloneqq V_+(\mathfrak{a})(K)$. We say X is **integral** if $X\left(\overline{k}\right)$ is irreducible in the subspace topology. We also define the **homogenous coordinate ring***

$$k[X]_{\text{hom}} \coloneqq k[X_0, ..., X_n]/\mathfrak{a}$$

*which is naturally a positive graded ring by (3.11.12). Define the **irrelevant ideal***

$$k[X]_+ \coloneqq \bigoplus_{d>0} k[X]_d$$

We may generalise (6.4.2) as follows.

Proposition 6.4.4 (Zero Loci in Projective Varieties)
*Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $A = k[X]_{\text{hom}}$ the coordinate ring. For any homogenous ideal $\mathfrak{b} \lhd A$ and field extension K/k define the **zero-locus** by*

$$V_+(\mathfrak{b})(K) \coloneqq \{(x_0 : ... : x_n) \in X(K) \mid f(x_0, ..., x_n) = 0 \quad \forall f \in \mathfrak{b}\}$$

Similarly for a subset $Y \subset X(K)$ define

$$I_+(Y) \coloneqq \{f \in k[X] \mid f_d(x) = 0 \quad \forall d \geq 0, x \in Y\}$$

Suppose $\pi : k[X_0, ..., X_n] \to k[X]$ is the quotient map. Then we have the following properties

$$V_+(\mathfrak{b})(K) = V_+(\pi^{-1}(\mathfrak{b}))(K) \cap X(K)$$
$$I_+(Y \cap X(K)) = \pi(I_+(Y)) \quad Y \subset \mathbb{P}^n_k(K)$$
$$I_+(V_+(\mathfrak{b})(K)) = \pi(I_+(V_+(\pi^{-1}(\mathfrak{b}))))$$

The pair $(V_+(-)(K), I_+)$ satisfies the same properties as (6.4.2). The image of $V_+(-)(K)$ forms the closed sets of a topology on $X(K)$ which coincides with the subspace topology from $\mathbb{P}^n_k(K)$. Further the Weak and Strong Nullstellensatz also hold

$$V_+(\mathfrak{b})\left(\overline{k}\right) = \emptyset \iff k[X]_+ \not\subset \sqrt{\mathfrak{b}} \iff \mathfrak{b} \text{ is irrelevant}$$

and

© David Rufino CC BY-NC-ND 4.0

$$\mathfrak{b} \text{ essential} \implies I_+\left(V_+(\mathfrak{b})\left(\overline{k}\right)\right) = \sqrt{\mathfrak{b}}$$

Further there is a dual isomorphism

$$\left\{\mathfrak{b} \lhd k[X]_{\text{hom}} \;\middle|\; \mathfrak{b} \text{ radical, homogenous and } A_+ \not\subset \mathfrak{b}\right\} \underset{I_+}{\overset{V_+}{\rightleftarrows}} \left\{Z \in X\left(\overline{k}\right) \;\middle|\; Z \neq \emptyset\right\}$$

Proposition 6.4.5 (Projective Subvariety)
*Let X be a projective variety, $\mathfrak{b} \lhd k[X]_{\text{hom}}$ an essential radical homogenous ideal. Then $V_+(\mathfrak{b})(-)$ is a **closed subvariety** which coincides with the projective variety $V_+(\pi^{-1}(\mathfrak{b}))$. There is a bijective correspondence between non-empty closed subsets of $X\left(\overline{k}\right)$ and closed subvarieties of X.*

Conversely if X', X are two projective varieties such that $X'\left(\overline{k}\right) \subset X\left(\overline{k}\right)$ then X' may be expressed as a closed subvariety.

Proof. There is a canonical graded surjective k-algebra homomorphism $\pi : k[X_0, ..., X_n] \to k[X]_{\text{hom}}$ by (3.11.12). Then $\pi^{-1}(\mathfrak{b})$ is graded (3.11.10) and radical (3.3.50) g). Also by definition of the quotient grading on π we find $\pi^{-1}(k[X]_+) = k[X_0, ..., X_n]_+$ so that $\mathfrak{b}$ essential $\implies \pi^{-1}(\mathfrak{b})$ is essential (3.12.7). The bijective correspondence exists by (6.4.4).

If $X' = V_+(\mathfrak{b}')$, $X = V_+(\mathfrak{b})$ are two projective varieties such that $X'\left(\overline{k}\right) \subset X\left(\overline{k}\right)$ then by the dual isomorphism we have $\mathfrak{b} \subset \mathfrak{b}'$. Therefore X' may be represented as the closed subvariety $V_+(\pi(\mathfrak{b}'))$ ∎

Proposition 6.4.6 (Principal Open Sets)
Let $X \subset \mathbb{P}^n_k$ be a projective variety. The Zariski topology on $X(K)$ has a base consisting of subsets of the form

$$D_+(f)(K) \coloneqq X(K) \setminus V_+((f)) = \{(x) \in X(K) \mid f_d(x) \neq 0 \text{ some } d > 0\}$$

where $f \in k[X]$ is a homogenous form. Note that $(x) = (y) \implies (f_d(x) = 0 \iff f_d(y) = 0)$ so the second form is well-defined.

Proof. A generic open set $U = X(K) \setminus V_+(\mathfrak{b})(K)$ where $\mathfrak{b}$ is a homogenous ideal of $k[X]$. Given $(x) \in U$ there is some $f \in \mathfrak{b}$ and $d > 0$ such that $f_d(x) \neq 0$. Therefore $D_+(f_d) \subset U$ is an open neighbourhood of (x). ∎

Proposition 6.4.7
Let $X = V_+(\mathfrak{b})$ be a projective variety and $f, g \in k[X]$ essential homogenous forms. Then

$$D_+(g) \subset D_+(f) \iff f \mid g^N \text{ some } N > 0$$

Proof.

$$D_+(g) \subset D_+(f) \iff V_+((f)) \subset V_+((g)) \iff \sqrt{(g)} \subset \sqrt{(f)} \iff f \mid g^N$$

∎

Proposition 6.4.8 (Irreducible Subsets)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $\mathfrak{b} \lhd k[X]$ a homogenous radical ideal. Then a closed subvariety $V_+(\mathfrak{b}) \subset X$ is integral iff $\mathfrak{b}$ is prime.

In particular $\mathbb{P}^n_k\left(\overline{k}\right)$ is integral.

Proof. Let $Y \coloneqq V_+(\mathfrak{b})$. Suppose that Y is not irreducible then $Y \subseteq V_+(\mathfrak{c}) \cup V_+(\mathfrak{d})$ is a non-trivial decomposition into closed subsets for $\mathfrak{c}, \mathfrak{d}$ essential homogenous radical ideals for which $Y \not\subset V_+(\mathfrak{c}) \implies \mathfrak{c} \not\subset \mathfrak{b}$ and similarly $\mathfrak{d} \not\subset \mathfrak{b}$ (see (4.1.61)). Choose $f \in \mathfrak{c} \setminus \mathfrak{b}$ and $g \in \mathfrak{d} \setminus \mathfrak{b}$. Then fg is zero on Y whence $fg \in \mathfrak{b}$. This shows that $\mathfrak{b}$ is not prime.

Conversely suppose $Y \subset V_+(\mathfrak{c}) \cup V_+(\mathfrak{d}) = V_+(\mathfrak{c} \cap \mathfrak{d})$ then by (6.4.2) $\mathfrak{c}\mathfrak{d} \subset \mathfrak{c} \cap \mathfrak{d} \subset \mathfrak{b}$. By (3.3.39) we have $\mathfrak{c} \subset \mathfrak{b}$ or $\mathfrak{d} \subset \mathfrak{b}$, whence $Y \subset V_+(\mathfrak{c})$ or $Y \subset V_+(\mathfrak{d})$. Therefore we conclude that Y is irreducible by (4.1.61).

Therefore $\mathbb{P}^n_k\left(\overline{k}\right)$ is irreducible by the special case both $\mathfrak{a}$ and $\mathfrak{b}$ being the zero ideals. ∎

Proposition 6.4.9 (Dual Isomorphism between Closed Sets and Homogenous Ideals)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $A = k[X]$ the coordinate ring. Under the dual isomorphism

$$\left\{\mathfrak{b} \lhd A \;\middle|\; \mathfrak{b} \text{ radical homogenous and } A_+ \not\subset \mathfrak{b}\right\} \underset{I_+}{\overset{V_+}{\rightleftarrows}} \left\{Z \subset X\left(\overline{k}\right) \text{ closed} \;\middle|\; Z \neq \emptyset\right\}$$

we have the following correspondence

- *irreducible sets correspond to prime ideals*

© David Rufino CC BY-NC-ND 4.0

- *irreducible components (of $V_+(\mathfrak{b})$) correspond to minimal prime ideals (of $\mathfrak{b}$)*

Proposition 6.4.10 (Decomposition into Irreducible Components)
Let $X = V_+(\mathfrak{a})$ be a projective variety then the topological space $X\left(\overline{k}\right)$ is Noetherian. Furthermore every closed subset Z has finitely many irreducible components Z_i and the decomposition

$$Z = Z_1 \cup ... \cup Z_n$$

is the unique incomparable decomposition into irreducible closed subsets.

If $Z = V_+(\mathfrak{b})$ then the irreducible components correspond to precisely the minimal prime ideals over $\mathfrak{b}$.

Proof. $X\left(\overline{k}\right)$ is Noetherian by using the order isomorphism in (6.4.9) and the fact $k[X]$ is Noetherian. The result then follows from (4.1.76). ∎

6.4.2 Affine Charts

References:

- Algebraic Curves [Ful08] (Chap. 4.3)
- Algebraic Geometry [Har13] (Chap. I §2)

For every $i = 0, ..., n$ the subset of $\mathbb{P}^n_k(K)$ consisting of points of the form

$$\{(x_0 : ... : x_n) \mid x_i \neq 0\}$$

may be naturally identified with $\mathbb{A}^n_k(K)$. The purpose of this section is to make this notion precise.

Lemma 6.4.11
There is an isomorphism of rings

$$\begin{aligned} \theta_i^\sharp : k[X_0, ..., X_n]_{(X_i)} &\longleftrightarrow k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n] \\ \frac{X_j}{X_i} &\longleftrightarrow x_j \end{aligned}$$

Proof. The algebra homomorphism $\theta_i^\sharp$ exists by (3.7.6) and (3.5.5), restricting to the subring $k[X_0, ..., X_n]_{(X_i)}$. The inverse algebra homomorphism exists by (3.7.6), sending $x_j \to \frac{X_j}{X_i}$. We may verify directly that these are mutually inverse. ∎

Lemma 6.4.12 ((De-)Homogenisation)
Fix n and $0 \leq i \leq n$. For $G \in k[X_0, ..., X_n]$ homogenous and $F \in k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ define

$$\begin{aligned} G^a &:= G(x_0, ..., x_{i-1}, 1, x_{i+1}, ..., x_n) \\ F^h &:= X_i^{\deg(F)} F\left(\frac{X_0}{X_i}, ..., \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, ..., \frac{X_n}{X_i}\right) \\ &= \sum_{j=0}^{\deg F} X_i^{\deg(F)-j} F_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n) \end{aligned}$$

Then

a) $(FF')^h = F^h F'^h$ and $(GG')^a = G^a G'^a$
b) $F^{ha} = F$ and $X_i^k G^{ah} = G$ where $k := v_{X_i}(G)$ is the smallest power of X_i appearing in G.
c) For $\mathfrak{b} \lhd k[X_0, ..., X_n]$ a homogenous ideal the set $\mathfrak{b}^a := \{G^a \mid G \in \mathfrak{b}\}$ is an ideal.

For $\mathfrak{a} \lhd k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ define the homogenous ideal

$$\mathfrak{a}^h := \left\langle \{F^h \mid F \in \mathfrak{a}\} \right\rangle$$

Then we also have the identities

d) $\mathfrak{b} \subset \mathfrak{b}^{ah}$ with equality when $\mathfrak{b}$ is radical and $\mathfrak{b} \subset \mathfrak{p}$ minimal $\implies X_i \notin \mathfrak{p}$
e) $\mathfrak{a} = \mathfrak{a}^{ha}$
f) $\mathfrak{b}$ prime and $X_i \notin \mathfrak{b} \implies \mathfrak{b}^a$ prime.
g) $\mathfrak{a}$ prime $\implies \mathfrak{a}^h$ prime
h) $\mathfrak{a}$ radical $\implies \mathfrak{a}^h$ radical

© David Rufino CC BY-NC-ND 4.0

Proof. Mostly tedious verification

a) Using the notation of (6.4.11) we see that $F^h = X_i^{\deg(F)}\left(\theta_i^\sharp\right)^{-1}(F)$ so evidently $(FF')^h = F^h F'^h$ because $\deg(FF') = \deg(F) + \deg(F')$.

   The second relation follows immediately from (6.4.11).

b) The first identity is clear. We may write

$$G = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} X_i^{\deg(G)-j} G_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n)$$

   where $\deg\left(G_j\right) = j$ and $v_{X_i}(G)$ is the smallest power of X_i appearing in G. So

$$G^a = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} G_j(x_0, ..., x_{i-1}, x_{i+1}, ..., x_n)$$

$$G^{ah} = \sum_{j=0}^{\deg(G)-v_{X_i}(G)} X_i^{\deg(G)-v_{X_i}(G)-j} G_j(X_0, ..., X_{i-1}, X_{i+1}, ..., X_n)$$

   whence $G = X_i^{v_{X_i}(G)} G^{ah}$.

c) This is evident because $(-)^a$ is a ring homomorphism.

d) The first part is evident from b). Suppose $G \in \mathfrak{b}^{ah}$ then

$$G = \sum_{j=1}^{r} \lambda_j G_j^{ah} \Longrightarrow X_i^k G = \sum_{j=1}^{r} \lambda_j X_i^{k_j} G_j$$

   for $G_j \in \mathfrak{b}$. Then $X_i^k G \in \mathfrak{b} \Longrightarrow G \in \mathfrak{b}$ as we assumed $X_i \notin \mathfrak{b}$.

e) Trivially $F \in \mathfrak{a} \Longrightarrow F^h \in \mathfrak{a}^h \Longrightarrow F^{ha} = F \in \mathfrak{a}^{ha}$. Conversely any $G \in \mathfrak{a}^h$ is of the form

$$G = \sum_j \lambda_j F_j^h \Longrightarrow G^a = \sum_j \lambda_j^a F_j^{ha} = \sum_j \lambda_j^a F_j \in \mathfrak{a}^{ha}$$

   for some $\lambda_i \in k[X_0, ..., X_n]$.

f) Suppose $FF' \in \mathfrak{b}^a \Longrightarrow F^h F'^h \in \mathfrak{b}^{ah} = \mathfrak{b} \Longrightarrow F^h \in \mathfrak{b}$ or $F'^h \in \mathfrak{b} \Longrightarrow F \in \mathfrak{b}^a$ or $F' \in \mathfrak{b}^a$.

g) Suppose $GG' \in \mathfrak{a}^h$ then $GG' = \sum_j \lambda_j F_j^h$ for $F_j \in \mathfrak{a}$. Then $G^a G'^a = \sum_j \lambda_j^a F_j^{ha} = \sum_j \lambda_j^a F_j \in \mathfrak{a}$. Therefore say $G^a \in \mathfrak{a} \Longrightarrow G^{ah} \in \mathfrak{a}^h \Longrightarrow G = X_i^k G^{ah} \in \mathfrak{a}^h$.

h) Suppose $G^n \in \mathfrak{a}^h$, then $(G^a)^n \in \mathfrak{a}^{ha} = \mathfrak{a}$ whence $G^a \in \mathfrak{a}$ and $G^{ah} \in \mathfrak{a}^h \Longrightarrow G \in \mathfrak{a}^h$

■

Proposition 6.4.13 (Projective Space has Affine Covering)
Consider subsets $D_+(X_i)(K) = U_i(K) \subset \mathbb{P}_k^n(K)$ of the form

$$D_+(X_i)(K) = U_i(K) \coloneqq \{(x_0 : ... : x_n) \mid x_i \neq 0\} \quad i = 0, ..., n$$

These are open in the Zariski topology. Furthermore under the subspace topology U_i is homeomorphic to $\mathbb{A}_k^n(K)$, with an explicit homeomorphism given by

$$\theta_i : \mathbb{A}_k^n(K) \to U_i$$
$$(x_0, ..., x_{i-1}, x_{i+1}, ..., x_n) \to [x_0 : ... : 1 : ... : x_n]$$

For $F \in k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$ and $(x) \in U_i$ we have

$$F(x) = 0 \Longleftrightarrow F^h(\theta_i(x)) = 0$$

and similarly for $G \in k[X_0, ..., X_n]$ we have

$$G(\theta_i(x)) = 0 \Longleftrightarrow G^a(x) = 0$$

Further we have

a) $\theta_i\left(V^{\mathrm{aff}}(\mathfrak{a})\right) = V_+\left(\mathfrak{a}^h\right) \cap U_i$ for $\mathfrak{a} \lhd k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]$
b) $\theta_i\left(V^{\mathrm{aff}}(\mathfrak{b}^a)\right) = V_+(\mathfrak{b}) \cap U_i = V_+\left(\mathfrak{b}^{ah}\right) \cap U_i$ for $\mathfrak{b} \lhd k[X_0, ..., X_n]$ homogenous
c) $I^{\mathrm{aff}}(Y)^h = I_+(\theta_i(Y))$ for $Y \subset U_i$

© David Rufino CC BY-NC-ND 4.0

Proof. This follows largely from (6.4.12). The relations show that $\theta_i(D(F)) = D_+(F^h)$ and $\theta_i^{-1}(D_+(G)) = D(G^a)$, so that θ_i is a homeomorphism as required.

a) $x \in \theta_i(V^{\text{aff}}(\mathfrak{a})) \Longrightarrow x \in U_i$ and $F(\theta_i^{-1}(x)) = 0 \quad \forall F \in \mathfrak{a} \Longrightarrow x \in U_i$ and $F^h(x) = 0 \quad \forall F \in \mathfrak{a} \Longrightarrow G(x) = 0 \quad \forall G \in \mathfrak{a}^h \Longrightarrow x \in V_+(\mathfrak{a}^h) \cap U_i$. Conversely $x \in V_+(\mathfrak{a}^h) \cap U_i \Longrightarrow G(x) = 0 \quad \forall G \in \mathfrak{a}^h \Longrightarrow F(\theta_i^{-1}(x)) = 0 \quad \forall F \in \mathfrak{a}^{ha} = \mathfrak{a} \Longrightarrow x \in \theta_i(V^{\text{aff}}(\mathfrak{a}))$

b) $\theta_i(V^{\text{aff}}(\mathfrak{b}^a)) = V_+(\mathfrak{b}^{ah}) \cap U_i$. It remains to show that $V_+(\mathfrak{b}) \cap U_i = V_+(\mathfrak{b}^{ah}) \cap U_i$. As (6.4.12) $\mathfrak{b} \subset \mathfrak{b}^{ah}$ then $V_+(\mathfrak{b}^{ah}) \cap U_i \subset V_+(\mathfrak{b}) \cap U_i$. Conversely if $x \in V_+(\mathfrak{b}) \cap U_i$ then $G(x) = 0 \Longrightarrow x_i^k G^{ah}(x) = 0 \Longrightarrow G^{ah}(x) = 0$ because $x_i \neq 0$. This shows $x \in V_+(\mathfrak{b}^{ah}) \cap U_i$

c) Suppose $G \in I^{\text{aff}}(Y)^h$ then $G = \sum_j \lambda_j F_j^h$ for $F_j \in I^{\text{aff}}(Y)$ and $\lambda_j \in k[X_0, ..., X_n]$. Then $F_j(x) = 0 \Longrightarrow F_j^h(\theta_i(x)) = 0 \Longrightarrow G(\theta_i(x)) = 0$. Therefore we conclude $G \in I_+(\theta_i(Y))$. Conversely if $G \in I_+(\theta_i(Y))$ then $G^a(x) = 0$ for all $x \in Y$ and $G^a \in I^{\text{aff}}(Y) \Longrightarrow G^{ah} \in I^{\text{aff}}(Y)^h \Longrightarrow G = X_i^k G^{ah} \in I^{\text{aff}}(Y)^h$.

■

Lemma 6.4.14
Let $X = V_+(\mathfrak{b})$ be a projective variety. Then $X \cap U_i \neq \emptyset \iff X_i \notin \mathfrak{b} \iff \overline{X_i} \neq 0$.

Proof. Observe $X \cap U_i = \emptyset \iff X \subset V_+(X_i) \iff X_i \in I_+(X) = \mathfrak{b}$. ■

Corollary 6.4.15
Let $X = V_+(\mathfrak{b}) \subset \mathbb{P}_k^n$ be a projective variety and for each affine chart $U_i \subset \mathbb{P}_k^n$ consider the affine variety $X^a := V^{\text{aff}}(\mathfrak{b}^a) \subset \mathbb{A}_k^n$. Then there is a natural homeomorphism

$$\theta_i : X^a(K) \xrightarrow{\sim} D_+(\overline{X_i}) = X(K) \cap U_i(K)$$

for all algebraic extensions K/k.

Proposition 6.4.16 (Projective Closure)
Identify $\mathbb{A}_k^n(\overline{k})$ with $U_i \subset \mathbb{P}_k^n(\overline{k})$. Let $Y \subset U_i$ be a subset. Then we have the following expression for topological closure

$$\overline{Y} = V_+\left(I^{\text{aff}}(Y)^h\right)$$
$$\overline{Y} \cap U_i = \text{cl}_{U_i}(Y)$$

*We call $\overline{Y}$ the **projective closure** of Y.*

Proof. By (6.4.13).c) $I_+(Y) = I^{\text{aff}}(Y)^h \Longrightarrow \overline{Y} \overset{(6.4.4)}{=} V_+ I_+(Y) = V_+\left(I^{\text{aff}}(Y)^h\right)$.

Then $\text{cl}_{U_i}(Y) = \overline{Y} \cap U_i$ by (4.1.21) (or more directly $\overset{(6.1.2)}{=} V_+(I_+(Y)) \overset{(6.4.13).a)}{=} V_+\left(I^{\text{aff}}(Y)^h\right) \cap U_i = \overline{Y} \cap U_i$). ■

6.4.3 Galois Orbits

Proposition 6.4.17
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}_k^n$ be a projective variety and K/k a normal algebraic extension (e.g. $\overline{k}$). Then there is a well-defined group action

$$\text{Aut}(K/k) \times X(K) \to X(K)$$
$$(\sigma, (x_0 : ... : x_n)) \to (\sigma(x_0) : ... : \sigma(x_n))$$

which commutes with the affine charts

$$\sigma(\theta_i(x)) = \theta_i(\sigma(x)) \quad \forall x \in U_i(K) \cap X(K)$$

Write $X_0(K)$ for the set of equivalence classes under this action. The projection map

$$\pi : X(K) \to X_0(K)$$

is the Kolmogorov Quotient (4.1.36). In particular points $x, y \in X(K)$ are topologically indistinguishable iff they are Galois conjugate. Further closed (resp. open) subsets are Galois-equivariant.

Proof. First consider the case $X = \mathbb{P}_k^n$. If $(x_0 : ... : x_n) = (x_0' : ... : x_n')$ then $x_i = \lambda_i x_i'$ for $\lambda_i \in K^\star$. Then $\sigma(x_i) = \sigma(\lambda_i)\sigma(x_i')$ so $(\sigma(x_0) : ... : \sigma(x_n)) \sim (\sigma(x_0') : ... : \sigma(x_n'))$ and the action is well-defined. For $x \in X(K)$ write $\sigma(x)$. Then $\tau(\sigma(x)) = (\tau \circ \sigma)(x)$ so it is a group action.

For the general case if $F(x) = 0$ for $F \in k[X_0, ..., X_n]$ then $F(\sigma(x)) = 0$. Therefore $x \in V(\mathfrak{a})(K) \Longrightarrow \sigma(x) \in V(\mathfrak{a})(K)$ and the action is well-defined on $X(K)$.

© David Rufino CC BY-NC-ND 4.0

We need to show that two points $x, y \in X(K)$ are conjugate by $\mathrm{Aut}(K/k)$ iff they are topologically indistinguishable. Suppose $x = \sigma(y)$ and $x \in V(\mathfrak{b})$. Then evidently $y \in V(\mathfrak{b})$. Conversely if they are topologically indistinguishable then we may suppose $x, y \in U_i$. Then $\theta_i(x), \theta_i(y)$ are also indistinguishable, so by (6.1.9) they are Galois conjugate, and so $\theta_i(x) = \theta_i(\sigma(y))$. As θ_i is bijective we deduce $x = \sigma(y)$ as required. ∎

Proposition 6.4.18
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $L/K/k$ a tower of normal extensions. Then there is an injective map

$$\begin{array}{ccc} X(K) & \xhookrightarrow{i_{KL}} & X(L) \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} \\ X_0(K) & \overset{i_0}{\dashrightarrow} & X_0(L) \end{array}$$

Proof. In order for i_0 to be well-defined, we require that $\pi(x) = \pi(y) \implies \pi(i_{KL}(x)) = \pi(i_{KL}(y))$. Recall $\pi(x) = \pi(y) \implies x = \sigma(y)$ for some $\sigma \in \mathrm{Aut}(K/k)$. Then $i_{KL} \circ \sigma \in \mathrm{Mor}_k(K, L)$ and by (3.18.69) there exists $\hat{\sigma} \in \mathrm{Aut}(L/k)$ such that $\hat{\sigma}|_K := \hat{\sigma} \circ i_{KL} = i_{KL} \circ \sigma$. Therefore $\hat{\sigma}(i_{KL}(x)) = i_{KL}(y) \implies \pi(i_{KL}(x)) = \pi(i_{KL}(y))$.

To show i_0 is injective we need to show the converse, which follows from (3.18.71). ∎

6.4.4 Projective Varieties are Abstract Varieties

Definition 6.4.19 (Regular Function)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety and $U \subset X\left(\overline{k}\right)$ an open subset. We say a function

$$f : U \to \overline{k}$$

*is **regular** at $x \in U$ there is a neighbourhood V of x such that*

$$f(y) = \frac{g(y)}{h(y)} \quad \forall\, y \in V$$

*for $g, h \in k[X]$ homogenous polynomials of the same degree. We therefore define the **structure sheaf** as follows*

$$\mathcal{O}_X(U) := \left\{ f : U \to \overline{k} \;\middle|\; f \text{ regular at all } x \in U \right\}$$

The pair $(X, \mathcal{O}_X)$ is a space of functions.

Proposition 6.4.20 (Structure Sheaf of Regular Functions)
Let $X = V_+(\mathfrak{a}) \subset \mathbb{P}^n_k$ be a projective variety, $U \subset X\left(\overline{k}\right)$ open and $f : U \to \overline{k}$ a function. Consider the affine charts

$$\theta_i : X_i^a\left(\overline{k}\right) \xrightarrow{\sim} D_+\left(\overline{X_i}\right) = X\left(\overline{k}\right) \cap U_i \quad i = 0, ..., n$$

where $X_i^a = V^{\mathrm{aff}}(\mathfrak{a}^a)$ as in (6.4.15). Then the following are equivalent

a) f is regular (at x) in the sense of (6.4.19)
b) $f \circ \theta_i : \theta_i^{-1}(U) \to \overline{k}$ is regular (at $\theta_i^{-1}(x)$) in the sense of (6.1.25) whenever $U \cap U_i \neq \emptyset$ ($x \in U \cap U_i$)

In other words for each $i = 1, ..., n$ there is an isomorphism of spaces with functions

$$\theta_i : \left(X_i^a\left(\overline{k}\right), \mathcal{O}_{X_i^a}\right) \xrightarrow{\sim} \left(D_+\left(\overline{X_i}\right), \mathcal{O}_X\,|_{D_+\left(\overline{X_i}\right)}\right)$$

In particular $(X, \mathcal{O}_X)$ is an abstract variety.

Proof. $a) \implies b)$ Let $x := \theta(\hat{x})$. There exists an open nbhd V of x such that $f(y) = \frac{g(y)}{h(y)}$ for all $y \in V$. By definition $g = G + \mathfrak{a}$ and $h = H + \mathfrak{a}$ for $G, H \in k[X_0, ..., X_n]$ homogenous polynomials of the same degree, for which H doesn't vanish on V. Then $\hat{V} := \theta_i^{-1}(V)$ is a open nbhd of $\hat{x}$. For all $\hat{y} \in \hat{V}$ we have

$$(f \circ \theta_i)(\hat{y}) = f(\theta_i(\hat{y})) = \frac{G(\hat{y})}{H(\hat{y})} = \frac{G^a(y)}{H^a(y)}$$

$b) \implies a)$ Suppose $x \in U \cap U_i$. We have by definition an open nbhd $\hat{V}$ of $\theta_i^{-1}(x)$ and $G, H \in k[x_0, x_1, ..., x_{i-1}, x_{i+1}, ..., x_n]$ for which

$$(f \circ \theta_i)(\hat{y}) = \frac{G(\hat{y})}{H(\hat{y})} \quad \forall \hat{y} \in \hat{V}$$

© David Rufino CC BY-NC-ND 4.0

As θ_i is a homeomorphism, $V := \theta_i\left(\hat{V}_i\right)$ is an open nbhd of x. By (6.4.13) we have for $y := \theta_i(\hat{y})$

$$f(y) = \frac{G^h(y)}{H^h(y)}$$

which holds for all $y \in V$. ∎

We may generalise the dehomogenisation

Proposition 6.4.21 (Structure Sheaf on Basic Affines)
Let $X = V_+(\mathfrak{p})$ be an irreducible projective variety and suppose $X \cap U_i \neq \emptyset$. Then there is an isomorphism of k-algebras

$$\begin{array}{ccc} \mathcal{O}_X\left(D_+\left(\overline{X_i}\right)\right) & \xrightarrow[\sim]{(6.4.20)} & \mathcal{O}_{X\cap U_i}(X \cap U_i) \\ \vdots & & \sim \uparrow (6.1.26) \\ k[X]_{(\overline{X_i})} & \xrightarrow{\sim} & k[X \cap U_i] \end{array}$$

$$\frac{\overline{X_0}}{\overline{X_i}} \longleftrightarrow \overline{x_i}$$

and the dashed arrow is defined in the obvious way (and is therefore an isomorphism).

Proof. By (6.4.14) $X_i \notin \mathfrak{p}$. We first demonstrate the bottom map is well-defined and an isomorphism. Consider the following commutative diagram

$$\begin{array}{ccccc} & & k[X_0,...,X_n]_{(X_i)} & \xrightarrow[\sim]{\theta_i^\sharp} & k[U_i] \\ & \swarrow & \downarrow \pi_1 & & \downarrow \pi_2 \\ k[X_0,...,X_n]_{(X_i)}/\mathfrak{p}_{(X_i)} & \xrightarrow[(3.11.17)]{\sim} & k[X]_{(\overline{X_i})} & \xrightarrow{\sim} & k[X \cap U_i] \end{array}$$

where the isomorphism $\theta_i^\sharp$ is defined in (6.4.11).

It follows from (3.3.56) that π_1 is uniquely defined and has kernel $\mathfrak{p}_{(X_i)}$ which equals $\mathfrak{p}_{X_i} \cap k[X_0,...,X_n]_{(X_i)}$ by (3.11.15). Therefore we may regard π_1 as a quotient map.

We also claim that the diagonal map $\pi_2 \circ \theta_i^\sharp$ has kernel $\mathfrak{p}_{(X_i)}$. It is enough by (3.5.5).b) and the preceding comment to show that the composite map $\pi_2 \circ (-)^a : k[X_0,...,X_n] \to k[X \cap U_i]$ has kernel $\mathfrak{p}$. However this follows from (6.4.12) for $F \in \ker(\pi_2 \circ (-)^a) \implies F^a \in \mathfrak{p}^a \implies F^a = G^a \implies F^{ah} = G^{ah} \in \mathfrak{p}^{ah} = \mathfrak{p} \implies F^{ah} \in \mathfrak{p} \implies X_i^k F \in \mathfrak{p} \implies F \in \mathfrak{p}$.

The required map exists by (3.3.56), as π_1 is a quotient map. Further it is injective, surjective and therefore an isomorphism

The map $k[X]_{(\overline{X_i})} \to \mathcal{O}_X\left(D_+\left(\overline{X_i}\right)\right)$ is defined in the obvious way. It is clear the diagram commutes and so this map is an isomorphism. ∎

Proposition 6.4.22
Let X be a projective variety. Then it is separated.

6.4.5 Regular Morphisms

For the category of quasi-projective varieties we introduce a number of equivalent definitions of morphism. In light of (6.4.20) then b) shows a function is regular iff it is regular in all affine coordinates.

Proposition 6.4.23 (Regular Morphism)
Let X be a projective variety, $U \subset X\left(\overline{k}\right)$ an open subset, Y a quasi-projective variety and $\phi : U \to Y\left(\overline{k}\right)$ a function. Then the following are equivalent

*a) **pulls back regular functions** ϕ is continuous and determines a morphism of sheaves*

$$\begin{aligned} \mathcal{O}_Y &\to \phi_\star \mathcal{O}_U \\ f &\to f \circ \phi \end{aligned}$$

© David Rufino CC BY-NC-ND 4.0

b) ***affine projections are regular*** *There is an open cover $U = \bigcup_{i\in I} U_i$ such that $\phi(U_i)$ is a subset of an affine chart $W_{j(i)}$ and*

$$\pi_k \circ \theta_{j(i)}^{-1} \circ \phi|_{U_i} : U_i \to \overline{k}$$

is regular (6.4.19) for all $i \in I$ and $k = 1, ..., m$

c) ***locally homogenous*** *There is an open cover $U = \bigcup_{i\in I} U_i$ and families of homogenous polynomials $\{f_{i0}, ..., f_{im}\} \subset k[X]_{\text{hom}}$ of the same degree such that $U_i \subset D_+(f_{i0}) \cup ... \cup D_+(f_{im})$ and*

$$\phi(y) = (f_{i0}(y) : ... : f_{im}(y)) \quad \forall y \in U_i$$

*In this case we say ϕ is **regular** or a **regular morphism** of quasi-projective varieties.*

Proof. $b) \implies a)$ We may assume wlog that U_i is contained in an affine chart V_i. Therefore by definition and (6.4.20)

$$\theta_{j(i)}^{-1} \circ \phi \circ \theta_i \mid_{\theta_i^{-1}(U_i)} : \theta_i^{-1}(U_i) \to \mathbb{A}_k^m$$

is regular as a map of quasi-affine varieties. Therefore by (6.1.39) it is continuous. Whence $\phi|_{U_i}$ is continuous by (6.4.13), and ϕ is continuous by (4.1.27).

Suppose $f \in \mathcal{O}_Y(W)$ then it is enough to show for all $x \in \phi^{-1}(W)$ that $f \circ \phi$ is regular at x. Suppose $x \in U_i$ and replace W with $W \cap W_{j(i)}$. Then by definition $f \circ \theta_{j(i)} : \theta_{j(i)}^{-1}(W) \to \overline{k}$ is regular at $\theta_{j(i)}^{-1}(x)$. Whence by (6.1.40)

$$f \circ \phi \circ \theta_i = \left(f \circ \theta_{j(i)}\right) \circ \left(\theta_{j(i)}^{-1} \circ \phi \circ \theta_i\right)$$

is regular at $\theta_i^{-1}(x)$, and therefore by (6.4.20) $f \circ \phi$ is regular at x.

$a) \implies b)$ For $x \in U$ there is some affine W_j such that $\phi(x) \in W_j \implies x \in \phi^{-1}(W_j) =: V_x$. By (6.4.20) $\pi_k \circ \theta_j^{-1} : W_j \to \overline{k}$ is regular whence by assumption $\pi_k \circ \theta_j^{-1} \circ \phi|_{V_x}$ is regular. As every point has a nbhd V_x we may find the required open cover.

$b) \implies c)$ By definition there exists homogenous $f_{i0}, ..., f_{im}, g_{i0}, ..., g_{im} \in k[X]_{\text{hom}}$ such that

$$\phi(y) = \left(\frac{f_{i0}(y)}{g_{i0}(y)} : ... : 1 : ... : \frac{f_{im}(y)}{g_{im}(y)}\right) \quad \forall y \in U_i$$

Therefore we may define

$$h_{il} := f_{il} \prod_{k\neq l} g_{ik}$$

Then evidently

$$\phi(y) = (h_{i0}(y) : ... : h_{im}(y)) \quad \forall y \in U_i$$

$c) \implies b)$ For every $x \in U$ there exists some open nbhd V_x such that

$$\phi(y) = (f_0(y) : ... : f_m(y)) \quad \forall y \in V_x$$

Suppose that $f_j(x) \neq 0$ then replace V_x with $V_x \cap D_+(f_j)$. Then

$$\left(\pi_k \circ \theta_j^{-1} \circ \phi|_{V_x}\right)(y) = \frac{f_k(y)}{f_j(y)}$$

which is regular by definition. The family $\{V_x\}_{x\in U}$ is the corresponding open cover. ∎

Corollary 6.4.24
Let $\phi : X \to Y$ and $\psi : Y \to Z$ be regular morphisms of quasi-projective varieties. Then so is $\psi \circ \phi : X \to Z$.

Proof. This is immediate by (6.4.23).a). ∎

Corollary 6.4.25
A function $f : X\left(\overline{k}\right) \to \overline{k}$ is regular iff it coincides with a regular morphism $f : X\left(\overline{k}\right) \to D_+(X_0) \cong \mathbb{A}_k^1$. In particular a regular function is continuous with respect to the co-finite topology on $\overline{k}$.

Proof. This follows from (6.4.23).b). ∎

© David Rufino CC BY-NC-ND 4.0

Proposition 6.4.26
Let $X \subset \mathbb{P}^n_k$ be a projective variety, $U \subset X\left(\overline{k}\right)$ an open subset and $U = \bigcup_{i\in I} U_i$ an open cover. Suppose that we have $f_{i0}, ..., f_{im} \in k[X]_{\text{hom}}$ homogenous of the same degree such that

a) For every $i \in I$

$$U_i \subset D_+(f_{i0}) \cup ... \cup D_+(f_{im})$$

b) For every pair $U_i \cap U_{i'} \neq \emptyset$ we have

$$f_{ik'}(y)f_{i'k}(y) = f_{ik}(y)f_{i'k'}(y) \quad \forall y \in U_i \cap U_{i'}, \forall k, k' = 0, ..., m$$

Then there exists a unique regular morphism $f : U \to \mathbb{P}^m_k\left(\overline{k}\right)$ such that

$$f(x) = (f_{i0}(x) : ... : f_{im}(x)) \quad \forall x \in U_i$$

6.4.6 Local Ring

Proposition 6.4.27
Let $X = V_+(\mathfrak{b}) \subset \mathbb{P}^n_k$ be a projective variety and $W = V_+(\mathfrak{p})$ an irreducible subset. Then there is an isomorphism

$$\begin{aligned} k[X]_{(\mathfrak{p}/\mathfrak{b})} &\xrightarrow{\sim} \mathcal{O}_{X,W} \\ \frac{\overline{F}}{\overline{G}} &\to \left[\left(D\left(\overline{G}\right), \frac{F(-)}{G(-)}\right)\right] \end{aligned}$$

More precisely for any affine chart U_i such that $U_i \cap W \neq \emptyset$ there is a commutative diagram

$$\begin{array}{ccccc} k[U_i]_{\mathfrak{p}^a} & \xrightarrow[(3.5.45)]{\sim} & k[U_i \cap X]_{\mathfrak{p}^a/\mathfrak{b}^a} & \xrightarrow[(6.1.57)]{\sim} & \mathcal{O}_{U_i\cap X, U_i\cap W} \\ \wr\downarrow & & & & \downarrow\wr \\ & & & & (4.3.17) \\ k[X_0, ..., X_n]_{(\mathfrak{p})} & \xrightarrow[(3.11.18)]{\sim} & k[X]_{(\mathfrak{p}/\mathfrak{b})} & \dashrightarrow^{\sim} & \mathcal{O}_{X,W} \end{array}$$

Proof. We construct the left vertical map, and then the dashed arrow is uniquely defined.

As $U_i \cap W \neq \emptyset$ then $W \not\subset V(X_i) \implies X_i \notin \mathfrak{p}$. So by (6.4.12).d) $\mathfrak{p}^{ah} = \mathfrak{p}$. To complete the diagram we consider the mutually inverse maps

$$\begin{aligned} k[U_i] = k[x_0, ..., x_{i-1}, x_{i+1}, ..., x_n]_{\mathfrak{p}^a} &\to k[X_0, ..., X_n]_{(\mathfrak{p})} \\ \frac{F}{F'} &\to \frac{F\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)}{F'\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)} \\ \frac{G^a}{G'^a} &\leftarrow \frac{G}{G'} \end{aligned}$$

Note $F'^h \in \mathfrak{p} \implies F' = F'^{ha} \in \mathfrak{p}^a$. Therefore $F' \notin \mathfrak{p}^a \implies F'^h = X_i^{\deg(F)} F'\left(\frac{X_0}{X_1}, ..., \frac{X_n}{X_i}\right) \notin \mathfrak{p} \implies F'\left(\frac{X_0}{X_1}, ..., \frac{X_n}{X_i}\right) \notin \mathfrak{p}$. Similarly $G'^a \in \mathfrak{p}^a \implies G'^{ah} \in \mathfrak{p}^{ah} = \mathfrak{p} \implies G' = X_i^k G'^{ah} \in \mathfrak{p}$, so $G' \notin \mathfrak{p} \implies G'^a \notin \mathfrak{p}^a$. Finally

$$\frac{G^a\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)}{G'^a\left(\frac{X_0}{X_i}, ..., \frac{X_n}{X_i}\right)} = \frac{X_i^{-\deg(G)} G}{X_i^{-\deg(G')} G'} = \frac{G}{G'}$$

so the maps are mutually inverse. ∎

Corollary 6.4.28
Let X be an irreducible projective variety with $W = V_+(\mathfrak{p})$ an irreducible subvariety. Then there is a commutative diagram

© David Rufino CC BY-NC-ND 4.0

$$\begin{array}{ccc} k[U_i]_{\mathfrak{p}^a} & \hookrightarrow & k[U_i]_{(0)} \\ \downarrow \sim & & \downarrow \sim \\ k[X]_{(\mathfrak{p})} & \hookrightarrow & k[X]_{((0))} \\ \downarrow \sim & & \downarrow \sim \\ \mathcal{O}_{X,W} & \hookrightarrow & k(X) \end{array}$$

In particular $k(X)$ is the field of fractions for $\mathcal{O}_{X,W}$.

Proof. The vertical maps are from (6.4.27) and the horizontal maps are the obvious ones. As the top map is injective and corresponds to field of fractions we deduce that $k(X)$ is field of fractions for $\mathcal{O}_{X,W}$. ∎

6.4.7 Dimension

Proposition 6.4.29
Projective space $\mathbb{P}^n_k$ has dimension n. Furthermore it is non-singular, normal and regular.

Proof. Projective space is integral (6.4.8) and has an open subset isomorphic to $\mathbb{A}^n_k$. So we are done by (6.2.54) and (6.1.50).

The conditions are local (6.2.81) (3.23.10). ∎

Proposition 6.4.30
Let $X = V_+(\mathfrak{p})$ be an irreducible projective variety then $\dim k[X] = \dim X + 1$.

Proof. We have $\overline{X_i} \neq 0$ for some i as X is non-empty. Therefore

$$\dim k[X] \overset{(3.34.31)}{=} \dim k[X]_{\overline{X_i}} \overset{(3.12.8)}{=} \dim k[X]_{(\overline{X_i})}[T, T^{-1}] \overset{(3.9.3)}{=} \dim k[X]_{(\overline{X_i})}[T]_T \overset{(3.34.31)}{=} \dim k[X]_{(\overline{X_i})}[T] \overset{(3.34.29)}{=} \dim k[X]_{(\overline{X_i})} + 1$$

By (6.4.21) $\dim k[X]_{(\overline{X_i})} = \dim k[X \cap U_i] = \dim X \cap U_i$.

Finally this equals $\dim X$ (4.1.88). ∎

Proposition 6.4.31
Let X be a projective variety and $x \in X\left(\overline{k}\right)$. Then

$$\overline{\{x\}} = \left\{\sigma(x) \;\middle|\; \sigma \in \operatorname{Aut}\left(\overline{k}/k\right)\right\}$$

is an irreducible set of order $\left[k\left(\mathfrak{m}_{X,x}\right) : k\right]_s$. Further X is a symmetric space.

Proof. We have $x \in U_i$ for some affine chart then we may use (6.4.20) to reduce to the affine case (6.1.18). ∎

Corollary 6.4.32
Let X be a projective variety and $Z \subset X$ a closed subset. Then the following are equivalent

a) $\dim Z = 0$
b) *Z is finite*

More precisely

$$Z = \overline{\{x_1\}} \cup ... \cup \overline{\{x_n\}}$$

for a finite number of points $x_1, ..., x_n \in Z$.

Proof. This follows from (4.1.91) and (6.4.31). ∎

Proposition 6.4.33
Let $X = \mathbb{P}^n_k\left(\overline{k}\right)$ and $G \in k[X_0, ..., X_n]$ an irreducible, homogenous polynomial of positive degree. Then $\dim V_+(G) = n - 1$.

Proof. Let $Y := V_+(G)$ then by (...) we have $Y \neq \emptyset$. Choose an affine patch U_i such that $U_i \cap Y \neq \emptyset$. Then $Y \cap U_i = V(G^a) \neq \emptyset$ which implies G^a has positive degree. By (...) $\dim Y \cap U_i = n - 1$. ∎

© David Rufino CC BY-NC-ND 4.0

6.4.8 Valuative Completeness

Lemma 6.4.34
Let R be a valuation ring for K. Suppose $x_0, ..., x_n \in K$ not all zero, then there exists $0 \leq j \leq n$ such that $\frac{x_i}{x_j} \in R$ for all $i = 0, ..., n$.

Proof. Recall (3.24.17) that the abelian group $K^\star/R^\star$ is totally ordered by the condition $x \leq y \iff yx^{-1} \in R^\star$. Choose j for which $x_j \neq 0$ such that x_j is minimal under this ordering. ∎

Proposition 6.4.35 (Projective Varieties are Complete)
Let $X = V_+(\mathfrak{p}) \subset \mathbb{P}^n_k$ be an irreducible projective variety. Let $k \subset R \subset k(X)$ be a valuation ring. Then there exists a point $x \in X\left(\overline{k}\right)$ such that $\mathcal{O}_{X,x} \subset R$ and $\mathfrak{m}_R \cap \mathcal{O}_{X,x} = \mathfrak{m}_{X,x}$.

More precisely there exists a local homomorphism $\Phi : R \to \overline{k}$ such that $\Phi|_{\mathcal{O}_{X,x}} = \mathrm{ev}_x$.

Proof. Denote the homogenous coordinate ring by simply $k[X]$. In light of (6.4.28) we may assume that $k(X) = k[X]_{((0))}$ and $\mathcal{O}_{X,x} = k[X]_{\mathfrak{p}_x}$ with $\mathfrak{m}_{X,x} = \left(\mathfrak{p}_x k[X]_{\mathfrak{p}_x}\right)_{(0)}$.

By (...) we may assume that $k(\mathfrak{m}_R)/k$ is algebraic and therefore by (3.18.61) there exists a local k-algebra homomorphisms

$$\Phi : R \to k(\mathfrak{m}_R) \hookrightarrow \overline{k}$$

We may assume wlog that $X \cap U_0 \neq \emptyset$ and therefore $X_0 \notin \mathfrak{p}$ where $X_0, ..., X_n$ are the homogenous coordinates. Define $x_i := \frac{X_i}{X_0} \in k[X]_{((0))}$ for $i = 0, ..., n$. By (6.4.34) there is $1 \leq j \leq n$ such that $t_i := \frac{x_i}{x_j} = \frac{\overline{X}_i}{\overline{X}_j} \in R$ for all $i = 0, ..., n$.

Note that $t_j = 1 \notin \mathfrak{m}_R$ so $\Phi\left(t_j\right) \neq 0$ we have a well-defined projective point $x = [\Phi(t_0) : ... : \Phi(t_n)] \in \mathbb{P}^n_k\left(\overline{k}\right)$. For any homogenous polynomial $F \in k[X_0, ..., X_n]$ of degree d we have

$$\begin{aligned} F(t_0, ..., t_n) &= \overline{X}_j^d F\left(\overline{X}_0, ..., \overline{X}_n\right) \\ &= \overline{X}_j^d \overline{F} \end{aligned}$$

regarding $k(X)$ as a subring of $\mathrm{Frac}(k[X])$ and $\bar{\cdot}$ denoting the reduction map $k[X_0, ..., X_n] \to k[X]$. Further

$$F(\Phi(t_0), ..., \Phi(t_n)) = \Phi(F(t_0, ..., t_n))$$

If $F \in \mathfrak{p}$ then $\overline{F} = 0$ and therefore $F(x) = 0$. This shows that $x \in X\left(\overline{k}\right)$.

We require to show that $\mathcal{O}_{X,x} \subset R$ and $\Phi|_{\mathcal{O}_{X,x}} = \mathrm{ev}_x$, from which it immediately follows that $\mathfrak{m}_R \cap \mathcal{O}_{X,x} = \mathfrak{m}_{X,x}$

Consider $g, h \in k[X]$ homogenous of the same degree with $h \notin \mathfrak{p}_x$. We claim that $g/h \in R$. For there exists $G, H \in k[X_0, ..., X_n]$ homogenous such that

$$\begin{aligned} g &= G\left(\overline{X_0}, ..., \overline{X_n}\right) \\ &= \overline{X}_j^d G(t_0, ..., t_n) \\ h &= H\left(\overline{X_0}, ..., \overline{X_n}\right) \\ &= \overline{X}_j^d H(t_0, ..., t_n) \\ \frac{g}{h} &= \frac{G(t_0, ..., t_n)}{H(t_0, ..., t_n)} \end{aligned}$$

Evidently $G(t_0, ..., t_n) \in R$ since R is a k-algebra. Further by assumption

$$\begin{aligned} 0 &\neq h(\Phi(t_0), ..., \Phi(t_n)) \\ &= H(\Phi(t_0), ..., \Phi(t_n)) \\ &= \Phi(H(t_0, ..., t_n)) \end{aligned}$$

which means $H(t_0, ..., t_n) \notin \mathfrak{m}_R \implies H(t_0, ..., t_n) \in R^\star$. Using the expression above for $\frac{g}{h}$ shows it lies in R as required. Further

© David Rufino CC BY-NC-ND 4.0

$$\begin{aligned}\Phi\left(\frac{g}{h}\right) &= \frac{\Phi(G(t_0, ..., t_n))}{\Phi(H(t_0, ..., t_n))} \\ &= \frac{G(\Phi(t_0), ..., \Phi(t_n))}{H(\Phi(t_0), ..., \Phi(t_n))} \\ &= \frac{G(x)}{H(x)} \\ &= \frac{g}{h}(x)\end{aligned}$$

as required. ■

© David Rufino CC BY-NC-ND 4.0

6.5 Arithmetic Varieties

Remark 6.5.1
This section is rough working notes rather than a polished treatment, kept here for personal reference. Omitted from the HTML editions.

Proposition 6.5.2
Let $\mathfrak{o}$ be a ring with quotient field k. For an ideal $\mathfrak{a} \lhd \mathfrak{o}[X_1, ..., X_n]$ define the zero-locus

$$|X| := V(\mathfrak{a}) := \left\{ x \in \overline{k}^n \;\middle|\; f(x_1, ..., x_n) = 0 \quad \forall f \in \mathfrak{a} \right\}$$

and the coordinate ring $\mathfrak{o}[X] := \mathfrak{o}[X_1, ..., X_n]/\mathfrak{a}$. Then

$$|X| = V(\mathfrak{a}_k) \subset \overline{k}^n$$

$$\mathfrak{a}_k := \left\{ \sum_{i=1}^{n} \lambda_i f_i \in k[X_1, ..., X_n] \;\middle|\; f_i \in \mathfrak{a}, \lambda_i \in k \right\}$$

where $\mathfrak{a}_k \lhd k[X_1, ..., X_n]$. For every $f \in \mathfrak{o}[X_1, ..., X_n]/\mathfrak{a}$ the sets of the form

$$D(f) := \{x \in |X| \mid f(x) \neq 0\}$$

form a basis for the usual Zariski topology. For every principal open set $D(f) \subset |X|$ define

$$\mathcal{O}_X(D(f)) := \left\{ \phi : D(f) \to \overline{k} \;\middle|\; \phi(x) = \frac{g(x)}{f^r(x)} \forall x \in D(f), \text{some } g \in \mathfrak{o}[X], r \geq 0 \right\}$$

then $\mathcal{O}_X$ extends uniquely to sheaf of $\overline{k}$-valued functions with the obvious $\mathfrak{o}$-algebra structure.

© David Rufino CC BY-NC-ND 4.0

6.6 Scheme Notes

Remark 6.6.1
This section is rough working notes rather than a polished treatment, kept here for personal reference. Omitted from the HTML editions.

6.6.1 Schemes over $\mathbb{Z}$

Let A be a finitely-generated $\mathbb{Z}$-algebra, it may be expressed in the form $A = \mathbb{Z}[X_1, ..., X_n]/\mathfrak{a}$. For any $\mathbb{Q}$-algebra R the rational points are given by $\mathrm{AlgHom}_{\mathbb{Z}}(A, [R]_{\mathbb{Z}}) = \mathrm{AlgHom}_{\mathbb{Q}}(A \otimes_{\mathbb{Z}} \mathbb{Q}, R)$.

6.6.2 Closed Points vs Maximal Ideals

In the case of affine schemes there is a straightforward relationship between closed points and maximal ideals.

Proposition 6.6.2
Let $X = \mathrm{Spec}(A)$ be an affine scheme. Then the closed points of X correspond precisely to maximal ideals of A.

For a general scheme a closed point corresponds to a maximal ideal of all affine neighbourhoods.

Proposition 6.6.3
Let X be a scheme, $x \in X$ and $U \subset X$ an affine neighbourhood, with $U \xrightarrow{\sim} \mathrm{Spec}(A)$. Then if x is a closed in X we have x is closed in U and corresponds to a maximal ideal of A.

Remark 6.6.4
However the converse may not hold – there may be maximal ideals of affines subsets which do not correspond to a closed point. As an extreme example it's possible to construct a scheme with no closed points at all. Consider a local ring A with the following ideal structure

$$0 \subsetneq \mathfrak{p}_0 \subsetneq ... \subsetneq \mathfrak{p}_n \subsetneq ... \subsetneq \mathfrak{m} \subsetneq A$$

Then $\mathrm{Spec}(A) \setminus \{[\mathfrak{m}]\}$ is an open subscheme with no closed points.

More simply consider the spectrum of a discrete valuation ring $\mathrm{Spec}(A) = \{(0), \mathfrak{m}\}$. Then $\{(0)\}$ is not closed in $X := \mathrm{Spec}(A)$, but it is open (being the complement of $V(\mathfrak{m})$) and even isomorphic to $\mathrm{Spec}\big(A_{(0)}\big)$, and so (0) is closed in an affine neighbourhood.

However we may obtain some partial converses

Proposition 6.6.5
Let X be a quasi-compact scheme, $U \subset X$ an affine open neighbourhood and $x \in U$ a closed point. Then there exists $y \in \overline{\{x\}}$ such that y is closed.

Proposition 6.6.6
Let X be a integral scheme locally of finite type over k. Then if $x \in U$ is closed for U affine, it is also closed in X.

Proof. By assumption we may write $X = \bigcup_{i \in I} V_i$ where V_i are affine (as finite type $\implies$ quasi-compact). Then by (4.1.21)

$$\mathrm{cl}_X(\{x\}) \cap V_i = \mathrm{cl}_{V_i}(\{x\})$$

whenever $x \in V_i$ (and it is empty otherwise). Therefore it is sufficient to show that $x \in V_i \implies x$ closed in V_i.

So we may consider the following situation: $U = \mathrm{Spec}(A)$ and $V = \mathrm{Spec}(B)$ are affine open neighbourhoods of x, for A, B finitely generated k-algebras, where additionally x is closed in U. Then by (6.6.2) x corresponds to a maximal ideal $\mathfrak{m}$ of A, and a prime ideal of $\mathfrak{p}$ of B. As these are both stalks of X we have a local k-algebra isomorphism $A_{\mathfrak{m}} \xrightarrow{\sim} B_{\mathfrak{p}}$, and therefore also of residue fields $k(\mathfrak{m}) \xrightarrow{\sim} k(\mathfrak{p})$. Therefore by following lemma we have x is closed in V as required. ∎

Lemma 6.6.7
Let A be a finitely-generated k-algebra and $\mathfrak{p}$ a prime ideal. Then the following are equivalent

a) $\mathfrak{p}$ is maximal
b) $[\mathfrak{p}]$ is a closed point of $\mathrm{Spec}(A)$.
c) $k(\mathfrak{p})/k$ is algebraic
d) $k(\mathfrak{p})/k$ is finite

In particular the closed points of $\mathrm{Spec}(A)$ correspond precisely to the maximal ideals of A.

© David Rufino CC BY-NC-ND 4.0

Proof. First we observe that $k(\mathfrak{p})/k$ is always finitely generated, so $c) \iff d)$ is simply (3.18.44). Further $a) \iff b)$ is simply (6.6.2).

$d) \implies a)$ We have an inclusion $A/\mathfrak{p} \hookrightarrow k(\mathfrak{p})/k$. We deduce that $A/\mathfrak{p}$ is a finite-dimensional, integral k-algebra and therefore a field (3.34.56).

$a) \implies d)$ As A is finitely generated, so is $k(\mathfrak{p})/k$. By Zariski's Lemma (3.34.23) it is finite. ∎

© David Rufino CC BY-NC-ND 4.0

Bibliography

[**AM69**] M. Atiyah and I.G. McDonald. *Introduction to Commutative Algebra*. Westview Press, 1969.

[**ARV10**] Jiří Ad\'amek, Jiří Rosický, and Enrico M. Vitale. *Algebraic Theories: A Categorical Introduction to General Algebra*. Cambridge University Press, 2010.

[**BS81**] Stanley Burris and H.P. Sankappanavar. *A Course in Universal Algebra*. Springer, 1981.

[**Bir35**] Garrett Birkhoff. On the Structure of Abstract Algebras. *Mathematical Proceedings of the Cambridge Philosophical Society* 31(4), 433--454, 1935.

[**Bir40**] G. Birkhoff. *Lattice Theory*. American Mathematical Society, 1940.

[**Bor94**] Francis Borceux. *Handbook of Categorical Algebra 2: Categories and Structures*. Cambridge University Press, 1994.

[**Bou89**] Nicolas Bourbaki. *Algebra: Chapters 4-7*. Springer-Verlag, 1989.

[**Bou98a**] Nicolas Bourbaki. *Algebra I: Chapters 1-3*. Springer, 1998.

[**Bou98b**] N. Bourbaki. *Commutative Algebra: Chapters 1-7*. Springer, 1998.

[**Bum98**] D. Bump. *Algebraic Geometry*. World Scientific, 1998.

[**Car71**] Henri Cartan. *Differential Calculus*. London: Kershaw Publishing Company, 1971.

[**Che51**] C. Chevalley. *Introduction to the Theory of Algebraic Functions of One Variable*. American Mathematical Society, 1951.

[**Die11**] Jean Dieudonné. *Foundations of modern analysis*. Read Books Ltd, 2011.

[**Eme10**] Matthew Emerton. Jacobson Rings, 2010.

[**For81**] Otto Forster. *Lectures on Riemann Surfaces*, 1981.

[**Fre64**] Peter Freyd. *Abelian Categories: An Introduction to the Theory of Functors*. Harper & Row, 1964.

[**Ful08**] William Fulton. Algebraic Curves. , 2008.

[**Gro57**] Alexander Grothendieck. Sur quelques points d'algèbre homologique. *Tôhoku Mathematical Journal* 9(2), 119--221, 1957.

[**Gro60**] Alexander Grothendieck. Éléments de géométrie algébrique : {I.} {Le} langage des schémas. *Publications Mathématiques de l'IHÉS* 4, 5--228, 1960.

[**Gro64**] Alexander Grothendieck. Éléments de géométrie algébrique : {IV.} {Étude} locale des schémas et des morphismes de schémas, {Première} partie. *Publications Mathématiques de l'IHÉS* 20, 5--259, 1964.

[**Hal17**] P.R. Halmos. *Naive Set Theory*. Dover Publications, 2017.

[**Har13**] Robin Hartshorne. *Algebraic Geometry*. Springer New York, 2013.

[**Hei17**] Katharina Heinrich. Some remarks on biequidimensionality of topological spaces and Noetherian schemes. *Journal of Commutative Algebra* 9(1), 49--63, 2017.

[**Hol11**] Audun Holme. *A royal road to algebraic geometry*. Springer-Verlag New York, 2011.

[**Kap74**] Irving Kaplansky. *Commutative Rings*. The University of Chicago Press, 1974.

[**Kel71**] John L Kelley. *General Topology*. Springer, 1971.

[**Lan11**] Serge Lang. *Algebra*. Springer New York, 2011.

[**Lan12**] Serge Lang. *Real and functional analysis*. Springer Science & Business Media, 2012.

[**Lan19**] Serge Lang. *Introduction to Algebraic Geometry*. Dover Publications, 2019.

[**Lan72**] Serge Lang. *Differential manifolds*. Springer, 1972.

[**Lan82**] Serge Lang. *Introduction to Algebraic and Abelian Functions*. Springer-Verlag, 1982.

[**Lan98**] Saunders Mac Lane. *Categories for the Working Mathematician*. Springer, 1998.

[**Liu06**] Qing Liu. *Algebraic Geometry and Arithmetic Curves*. Oxford University Press, 2006.

[**Mat70**] H. Matsumura. *Commutative Algebra*. Benjamin, 1970.

[**Mil14**] James Milne. Introduction to Commutative Algebra, 2014.

[**Mil17**] James Milne. Algebraic Geometry, 2017.

[**Mor12**] Patrick Morandi. *Field and Galois theory*. Springer Science & Business Media, 2012.

[**Mum99**] David Mumford. *The red book of varieties and schemes : includes the Michigan Lectures (1974) on Curves and their Jacobians*. New York: Springer, 1999.

[**Nag75**] M. Nagata. *Local Rings*. R.E. Krieger Publishing Company, 1975.

[**Neu99**] Jürgen Neukirch. *Algebraic Number Theory*. Berlin: Springer-Verlag, 1999.

[**Rey11**] Manny Reyes. Krull dimension less or equal than transcendence degree?, 2011. Math Overflow.

© David Rufino CC BY-NC-ND 4.0

[**Rie16**] Emily Riehl. *Category Theory in Context*. Dover Publications, 2016.

[**Rom05**] S. Roman. *Field Theory*. Springer New York, 2005.

[**Ser55**] Jean-Pierre Serre. Faisceaux algébriques cohérents. *Annals of Mathematics* 61(2), 197--278, 1955.

[**Ser59**] Jean-Pierre Serre. *Groupes algébriques et corps de classes*. Hermann, 1959.

[**Ser79**] Jean-Pierre Serre. *Local Fields*. New York: Springer-Verlag, 1979.

[**Sha00**] R.Y. Sharp. *Steps in Commutative Algebra*. Cambridge University Press, 2000.

[**Sha94**] Igor Shafarevich. *Basic algebraic geometry*. Springer-Verlag New York, 1994.

[**Shi55**] Goro Shimura. Reduction of Algebraic Varieties with Respect to a Discrete Valuation of the Basic Field. *American Journal of Mathematics* 77(1), 134--176, 1955.

[**Stacks**] The Stacks Project Authors. Stacks Project, 2015.

[**Sti09**] Henning Stichtenoth. *Algebraic Function Fields and Codes*. Springer Berlin Heidelberg, 2009.

[**Tat68**] John Tate. Residues of differentials on curves. *Annales scientifiques de l'École Normale Supérieure* 1(1), 149--159, 1968.

[**Wae91**] B. L. van der Waerden. *Algebra: Volume I*. Springer New York, 1991.

[**War13**] F. Warner. *Foundations of Differentiable Manifolds and Lie groups*. Springer-Verlag New York, 2013.

[**Wed16**] Torsten Wedhorn. *Manifolds, Sheaves, and Cohomology*. Springer, 2016.

[**Wil70**] Stephen Willard. *General Topology*. Dover Publications, 1970.

[**ZS75**] Oscar Zariski and Pierre Samuel. *Commutative Algebra I*. Springer New York, 1975.

[**ZS76**] Oscar Zariski and Pierre Samuel. *Commutative Algebra II*. Springer New York, 1976.

© David Rufino CC BY-NC-ND 4.0